

SGT HYDROEDGE | GREENX

Hydrogen Fuel Assist for Industrial Kilns

Improve firing efficiency. Cut fuel costs. Reduce emissions. Batch, continuous & cement kilns.

THE PROBLEM

Industrial kilns are among the most energy-intensive equipment in manufacturing. Whether firing ceramics, bricks, tiles, cement clinker, lime, or refractories, kilns consume massive quantities of fuel — natural gas, coal, diesel, biomass, or pet coke. Yet typical kiln energy efficiency ranges from just 5% to 20% for ceramic roller kilns and 50–65% for cement rotary kilns. The majority of input energy is lost through exhaust gases, radiation, and incomplete combustion. Every percentage point of efficiency gained translates to significant fuel savings at these operating scales.

THE SGT HYDROEDGE SOLUTION

SGT HydroEdge is the first Indian company to manufacture high-capacity HHO (oxyhydrogen) generators designed for industrial kiln applications. Our system generates hydrogen-oxygen gas on-demand through water electrolysis and introduces it into the kiln's combustion zone. Hydrogen's exceptional flame propagation speed (nearly 10x faster than natural gas), wide flammability range (4–75% in air), and catalytic properties enable more complete combustion of the primary fuel — whether gas, coal, biomass, diesel, or pet coke.

The result: your existing kiln burns fuel more completely, reaches and maintains firing temperatures more efficiently, and produces dramatically fewer emissions — across batch kilns, continuous roller kilns, tunnel kilns, and rotary cement kilns.

SGT'S PATENT-APPLIED TECHNOLOGY ADVANTAGE

SGT HydroEdge has developed proprietary, patent-applied HHO generator technology engineered specifically for industrial-scale combustion applications:

High HHO Yield	Patent-applied cell design delivers significantly higher hydrogen-oxygen gas output per unit of electrical input compared to conventional electrolyzers
Low Power Consumption	Optimised electrode geometry and electrolyte management reduce electricity consumption, ensuring the net energy benefit is maximised
Long Operational Life	Engineered for continuous industrial duty. Durable materials and design ensure years of reliable operation without performance degradation
Safety by Design	On-demand generation means zero hydrogen storage. Built-in pressure relief, flashback arrestors, and fail-safe controls. No explosive gas inventory on site.
Easy & Low-Cost Maintenance	Modular design for quick servicing. No specialised tools required. Consumables are water and standard electrolyte — inexpensive and widely available.

DIGITALLY CONNECTED — IOT & DIGITAL TWIN READY

Every SGT HydroEdge system is fully digitally connected.

Built-in IoT sensors continuously monitor system performance, HHO output, power consumption, and kiln operating parameters. This data feeds directly into GreenVision — SGT's proprietary AI-powered Digital Twin platform.

- **Real-time monitoring** of HHO system health, gas output, and kiln combustion efficiency gains
- **Emission tracking** — continuous measurement of CO₂, PM, CO, NO_x, and SO_x reductions from kiln operations
- **ESG-grade reporting** — automated reports documenting emission reductions for compliance, CBAM, and investor disclosures
- **Predictive maintenance** — AI detects performance degradation before failures, for both the HHO system and kiln equipment
- **Carbon credit documentation** — verified savings data supporting carbon credit generation and trading

WHAT GLOBAL RESEARCH PROVES

Researchers across the world have extensively studied hydrogen-enriched combustion in kilns across ceramic, cement, brick, and lime applications. The results consistently demonstrate efficiency improvements and emission reductions. Here is what the published research shows:

CERAMIC TILE KILNS — CONTINUOUS ROLLER KILNS

- Hydrogen-enriched combustion experiments on an industrial ceramic kiln with 102 burners demonstrated optimised energy efficiency of 63.13% for tile production, with reduced natural gas consumption at optimum hydrogen flow rates
- Ceramic roller kilns typically operate at very low energy efficiency (5–20%), with even modest combustion improvements deliver outsized fuel savings. Energy savings of up to 17% have been documented through improved combustion and heat recovery
- An EU-funded project (LIFE ZEF-tile) demonstrated that oxygen-enriched firing in ceramic tile kilns reduced CO emissions by 90%, NOx by 80%, and particulate matter by 70%, while consuming approximately 10% less fuel
- European kiln manufacturers report that advanced burner technology for tile kilns delivers 8–10% natural gas savings — HHO-assisted combustion achieves comparable or better results as a retrofit to existing kilns without burner replacement
- Tile quality testing across multiple hydrogen combustion trials confirmed that product quality remains consistent with conventional firing when combustion parameters are properly managed

BATCH KILNS — BRICKS, POTTERY & REFRACTORIES

- Batch kilns using firewood, coal, or gas experience significant incomplete combustion, especially during low-temperature drying stages where CO emissions are highest. HHO injection improves combustion completeness across all temperature stages
- The British Ceramic Confederation's feasibility programme demonstrated successful hydrogen firing across heavy clay (bricks), whitewares, and refractories, with products meeting technical specifications. Both batch and continuous kiln configurations were tested
- Research confirms that hydrogen's faster flame propagation and lower ignition energy enable more uniform temperature distribution inside batch kilns, reducing hot spots and cold zones that cause product defects and energy waste
- In brick kilns where firewood is the primary fuel, adding a combustion catalyst like HHO reduces the total firewood requirement and dramatically cuts CO emissions during the critical low-temperature drying phase where incomplete combustion is most severe

CEMENT KILNS — ROTARY KILNS

- CEMEX, the global cement manufacturer, has deployed hydrogen injection as a combustion catalyst across all its European cement kilns since 2021. The technology increased alternative fuel substitution rates by 8–10 percentage points and has cut thousands of tonnes of CO₂ annually
- Thermodynamic analysis shows that hydrogen-based fuel in cement production reduces CO₂ emissions by up to 44% compared to conventional coal firing, due to carbon-free combustion
- Hydrogen's high flame propagation speed stabilises combustion in cement kilns and enables higher use of difficult alternative fuels such as biomass and refuse-derived fuel — amplifying the decarbonisation benefit beyond direct hydrogen substitution
- Increasing oxygen concentration in kiln burning air from 21% to 27% improves clinker production by 7% and achieves more complete combustion with lower CO emissions — directly relevant to HHO injection which provides both hydrogen and oxygen simultaneously

LIME & CALCINATION KILNS

- Limak Cement pioneered hydrogen injection into a calciner (the lower-temperature pre-processing stage of cement production) using a hydrogen-rich mix, demonstrating that hydrogen combustion is effective not just in main kiln burners but across the full thermal process
- For lime kilns and calcination processes, hydrogen-enriched combustion provides the dual benefit of achieving required temperatures with less fuel while reducing the CO₂ generated from fuel combustion (distinct from the process CO₂ released during limestone decomposition)
- Research confirms that hydrogen's combustion characteristics are particularly well-suited for calcination, where maintaining uniform temperature distribution is critical for product quality and energy efficiency

These findings are based on research conducted by scientists and engineers at leading institutions and companies worldwide including Tsinghua University (China), British Ceramic Confederation (UK), Lucideon (UK), SACMI (Italy), Institute of Ceramic Technology ITC-AICE (Spain), CEMEX (Global), Hanson/Heidelberg Materials (UK), VDZ German Cement Works Association, Limak Cement (Turkey), and the EU LIFE programme. Research published in peer-reviewed journals indexed in ScienceDirect, MDPI, Elsevier, and Springer. Full references available on request.



SGT 3000 LPH – HHO Generator for Boilers

High-Efficiency | Modular | Made in India

- Output: Up to 3000 LPH HHO, on-demand & safe
- Technology: Liquid Cell – high efficiency, low maintenance
- Scalable: Modular design for higher HHO requirement
- Power: 10 kVA IGBT-based high-efficiency supply
- Operation: 24×7 continuous, up to 7 days non-stop
- Safety: Multiple electrical & process interlocks
- Retrofit: No boiler or fuel system modification

Designed, developed & manufactured by SGT HydroEdge in Pune, India

Application: Industrial boilers, Kilns & thermal systems

AT A GLANCE

2–9%

Fuel Efficiency Improvement

Up to 44%

CO₂ Emission Reduction

Up to 80%

NO_x Reduction

Up to 70%

PM Reduction

8–10%

Alt Fuel Use Increase

Kiln types supported	Batch kilns, continuous roller kilns, tunnel kilns, rotary cement kilns, lime kilns
Fuel types supported	Natural gas, coal, pet coke, diesel, biomass, firewood, alternative fuels
Industries served	Ceramics, tiles, bricks, cement, lime, refractories, glass, calcination, Special tools
Installation	Retrofit. No kiln modification. Minimal production disruption.
Hydrogen storage	None — on-demand generation. Zero storage risk.
Connectivity	IoT enabled. Integrates with SGT GreenVision Digital Twin.
Technology	Patent-applied. High yield. Low power. Long life.

Ready to Improve Your Kiln Efficiency & Cut Emissions?

Ask us for a no-obligation assessment of your kiln operations.

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SGT HydroEdge — Sustainability Infrastructure for Industry